ZEROGUARD: A Solver-Free Battery Safety Certificate for Autonomous Vehicles, from Robotaxis to Deep Space

Anchored Collapse and the geometry of the admissible input set

Abstract

Every autonomous vehicle that must operate unattended — a robotaxi that never sees a technician, a delivery drone flying forty sorties a day, an autonomous underwater vehicle six months into a deployment, a spacecraft that has not been touched since integration — runs into the same wall. Its battery must be pushed hard enough to be useful and held back enough to be safe, and the state that decides where that line falls is not measurable in the field. The standard answers are to identify the pack online, which requires excitation nobody can afford, or to solve a constrained optimisation every control step, whose iteration count a safety task slot cannot be sized around.

This paper develops a third answer and takes it across four physical media. We prove that when a system's constraints are monotone in a scalar input and one always-available input is safe, the admissible input set collapses to a single interval whose edges are found by bisection — in a *fixed* number of one-step model evaluations, with no optimiser, no convergence test, and no identification of the plant. The original form of this result assumes the safe input is zero, which is true of a cell being charged and false of essentially every vehicle doing its job: cut the current to a rotorcraft and it falls; cut it to a submersible and it loses the computer navigating it; cut it to a spacecraft in eclipse and the payload browns out.

We show that this apparent failure is a misdiagnosis. Zero *is* still safe for every temperature and voltage constraint on a discharging vehicle; what zero cannot do is serve the load, and that is a constraint of a different kind — a floor rather than a cap — for which the original theorem has no vocabulary. **Anchored Collapse** adds the floor family and a second bisection, raising the worst-case cost from 20 to 40 model evaluations, both fixed. It reduces to the published filter bit-for-bit on 30,000 states.

Across 16 vehicle platforms in four media — ground, air, water and vacuum — and 49 experiments totalling 470,923 simulated episodes, no plant constraint was breached while the certificate reported feasible. The admissible set was a single interval in all 64,000 dense scans. Worst-case latency across every platform was 123.9 μs, or 1.24 % of a 10 ms ISO 26262 task slot.

Four results contradicted the hypotheses they were written to test, and are reported as findings rather than repaired: the certificate *refuses* the calibrated cell for a rotorcraft before takeoff above 1.2C of hover draw; in a sealed pressure hull the binding constraint is plating at every water temperature, which is the one constraint the theorem enforces but does not certify; in deep space a *smaller* radiator delivers more charge, because the pack is cold-limited rather than heat-limited; and under an ice shelf closure leaves only 7 minutes before an actual breach against a 20-minute transit to safety, which makes closure the wrong trigger and converts a negative result into a quantitative design rule.

1 Introduction

The limiting resource for long-duration autonomy is not perception, planning, or control. It is *energy under certainty*: how hard a vehicle may push its battery when nobody is available to inspect it, and how confident anyone can be in the answer.

The gap between what a lithium-ion pack can physically deliver and what a battery management system will actually permit is large, and it is not there by accident. Fast charging drives three coupled failure modes — thermal runaway, over-potential, and lithium plating on the anode — whose onset depends on internal states that a production sensor suite does not measure. Series resistance roughly doubles over a pack's service life, plating risk climbs as temperature falls, and the safe current at end of life in cold weather is a fraction of the safe current when new and warm. A manufacturer facing that uncertainty does the only defensible thing and de-rates for the worst case, permanently.

For a privately owned car this is a mild inefficiency. For an autonomous fleet it is the economics of the whole enterprise. A robotaxi that charges twelve times a day accumulates a private car's decade of fast-charge stress in 3.0 weeks (§6); the de-rate it inherits was designed for a vehicle with a fundamentally different duty cycle. For a spacecraft it is worse still, because the de-rate is chosen before launch and can never be revisited.

1.1 Two standard answers, and why neither reaches the vehicle

The first is *identification*: estimate the pack's internal state online and de-rate against the estimate. Estimating resistance and capacity well requires excitation — current profiles rich enough to be informative — which an operational vehicle is not free to inject. Worse, the guarantee becomes conditional on the estimator having converged, which is a claim about a transient nobody audits.

The second is *constrained optimisation*: solve a small model-predictive control problem each step and let the solver enforce the constraints. This is the standard modern answer, and a good one, with a specific defect — a quadratic program's iteration count depends on its data. A safety-critical task slot is sized for the worst case, and if the worst case is unbounded the slot cannot be sized at all. In practice this is handled with iteration caps and fallback logic: by having a plan for when the guarantee does not hold.

1.2 The result

We take a third path, whose value is that it needs neither. The observation is geometric rather than numerical:

> If every constraint is monotone in a scalar input, and one input is known to be safe, then the set of admissible inputs is not a complicated region to be searched. It is a single interval, and its edges can be located by bisection in a fixed number of one-step model evaluations.

Bisection needs no optimiser, no convergence test, and — critically — no identification of the plant, because a monotone *bound* on the uncertain parameter suffices where a point estimate would be required. A datasheet end-of-life resistance figure is such a bound and is known before the vehicle ships.

The contribution of this paper is to establish that this result survives contact with real vehicles, which required repairing it first. Section 4 shows that the failure observed when the filter is placed on a hovering rotorcraft is not the failure it appears to be, and develops **Anchored Collapse**, which covers discharge as well as charge. Sections 6–9 then take it across four media — ground, air, water, vacuum — across 16 platforms and 49 experiments.

Contributions. **(i)** *Anchored Collapse* (§4.2), a generalisation covering constraints requiring the input to be *large* enough — what every vehicle's irreducible load imposes and what the original theorem cannot express — at a cost of 40 fixed evaluations rather than 20. **(ii)** A two-sided certificate and a *reserve* (§4.4): the gap between the edges is how much room the vehicle has left, which the one-sided version cannot report and which yields an abort rule for vehicles that cannot afford to find their limit empirically. **(iii)** A certificate that refuses *designs*, not just commands

(§7.1). **(iv)** Application across four media (§6–§9), including where the result does *not* hold and what that costs.

2 The autonomy energy bottleneck

Why a battery certificate is a bottleneck for autonomy rather than a component-level concern is worth being concrete about, because the four application sections that follow are chosen to span the ways it binds.

Ground fleets are duty-cycle limited. A robotaxi's economics are set by utilisation, and utilisation is set by how long it spends attached to a charger. The constraint is not the charger — §6 shows a 350 kW station is already past what the pack will accept — but how aggressively the pack may be driven at the temperature it happens to be, at the age it happens to be, with no technician in the loop. That is precisely the quantity a certificate computes.

Aerial autonomy is margin limited. A rotorcraft in hover draws a continuous 1–3C from its pack. There is no coasting, no regeneration, and no graceful degradation: an aircraft that runs out of admissible current does not slow down, it descends. Every kilogram of battery carried to buy margin is a kilogram not carried as payload, so the margin is contested directly against the mission.

Underwater autonomy is abort limited. An AUV under an ice shelf has no abort-to-surface. The vehicle must decide to turn back while turning back is still possible, using only what it can compute onboard, hours or days from any human. A certificate that is correct but late is indistinguishable from one that is wrong.

Space autonomy is calibration limited. A spacecraft's battery model is fixed at integration and cannot be updated in any meaningful sense afterwards — the downlink budget is not for battery parameters, and the round-trip light time to the outer planets is hours. Over a five-year mission the pack ages, takes radiation dose, and cycles tens of thousands of times against a parameter set that has not changed since before launch. **The property that makes a solver-free certificate merely convenient on a car is the property that makes it admissible at all on a spacecraft**: it requires no identification, so there is nothing to recalibrate.

These four are not a list of markets. They are four different ways for the same underlying guarantee to be the binding constraint, and a method that addresses one by exploiting its specifics would not transfer. What follows is deliberately one method, unmodified, in all four.

3 Background and related work

Set-invariance and predictive safety filters. The framework is control invariance [1]: a set is control-invariant if from every state in it some input keeps the next state inside. Control barrier functions [2] and predictive safety filters [3] enforce this by solving an optimisation each step; reference governors [4] and shielding [5] are related. Ours is a predictive safety filter whose optimisation has been replaced by a bisection, which is possible only because of the monotone structure below.

Monotone systems. The structural property doing the work is monotonicity [6; 7]: order-preserving dynamics let worst-case reachability be computed from extreme points rather than by searching an interior. Our use is narrower — monotonicity in a scalar *input* rather than in the state — but it is what turns a search into two bisections.

Batteries, degradation and learned policies. The reference model is Doyle–Fuller–Newman [8; 9] with Bernardi thermal coupling [10], implemented in PyBaMM [11] with the Chen [12] and O'Kane [13] parameter sets; plating is the dominant fast-charge degradation mode [14]. Online estimation of resistance and capacity is mature [15; 16] — and is exactly what this method deliberately does not require. Reinforcement learning has been applied to fast charging [17; 18], and constrained RL offers expectation-level guarantees [19; 20], which are the wrong shape for a constraint that must hold on every step of every episode. Our position is that the policy may be learned and the safety must not be.

4 Theory

Let a system have state x, scalar input $u \in [u_{\min}, u_{\max}]$, one-step map $x^+ = f(x, u, w)$, and a finite family of constraints $c_j(x, u) \le 0$ defining a safe set $\mathcal{S}$. Write $\mathcal{U}(x) = \{u : c_j(x, u) \le 0 \ \forall j\}$ for the admissible input set.

4.1 Null-Input Collapse

Proposition 1 (Null-Input Collapse)**.** *Suppose **(A1)** $u = 0$ is safe — from every $x \in \mathcal{S}$ the one-step map at $u = 0$ returns to $\mathcal{S}$ — and **(A2)** each c_j is non-decreasing in u. Then $\mathcal{U}(x) = [0, u^*(x)]$ is a single interval anchored at zero, $\mathcal{S}$ is forward-invariant under any policy filtered through it, and u^* is computed to precision ε in $\lceil \log_2(u_{\max}/\varepsilon) \rceil + 2$ one-step evaluations.*

Sketch. By (A2) each $\{u : c_j \le 0\}$ is a down-set; a finite intersection of down-sets in $\mathbb{R}$ is a down-set, hence an interval $[u_{\min}, u^*]$, which by (A1) contains 0. Bisection on a monotone predicate converges linearly with an iteration count depending only on the requested precision, not on the data. Forward invariance follows by induction: each filtered step lands in $\mathcal{S}$ and $u = 0$ remains available. □

Two things are worth isolating. First, no step requires knowing the plant's uncertain parameters — only evaluating c_j under a *bound* on them, and a conservative bound yields a conservative interval. Second, the cost is $O(\log 1/\varepsilon)$ *deterministic* evaluations, which is what a task slot can be sized around and a solver's iteration count is not.

4.2 Where it breaks, and why that reading is wrong

Placed on a quadrotor holding altitude, a filter built on Proposition 1 passes every one-step check for a mean of 22.3 consecutive steps and then violates in 100 % of episodes. The natural reading is that (A1) has failed: $u = 0$ is plainly not safe for an aircraft.

That reading is wrong, and the correction is the technical heart of this paper. Consider what $u = 0$ actually does to a discharging pack. No current flows, so there is no I^2R heating and the temperature relaxes toward ambient. Terminal voltage returns to open-circuit, so it cannot sag below a floor. State of charge does not fall. **Every temperature and voltage constraint is satisfied at $u = 0$, on a quadrotor in flight exactly as on a cell being charged.**

What $u = 0$ cannot do is fly the aircraft. The constraint that failed is of a different kind: a requirement that the input be *large* enough. Call these *floors*, as against the *caps* of Proposition 1. Every vehicle has at least one, because every vehicle has an irreducible load — the power it cannot stop drawing and remain a vehicle. Hover thrust for a rotorcraft. Hotel load for a submersible, whose computer and inertial navigation draw their watts whether the thrusters do or not, so that even a stationary AUV cannot command zero. Bus load for a spacecraft in eclipse. Contracted export power for a car doing vehicle-to-grid.

Charging is the special case where the irreducible load is zero — which is exactly why the original filter could anchor at zero without ever noticing it was making a choice.

Theorem 1 (Anchored Collapse)**.** *Partition the constraints into caps $\mathcal{C}$, each satisfied on $[u_{\min}, u_c]$, and floors $\mathcal{F}$, each satisfied on $[u_c, u_{\max}]$. Suppose **(A1′)** $u = u_{\min}$ is safe for every cap, from every $x \in \mathcal{S}$, and **(A2)** each constraint is monotone in u. Then*

$$\mathcal{U}(x) \;=\; [\,u_{\mathrm{lo}}(x),\; u_{\mathrm{hi}}(x)\,] \;=\; \Big[\,\max_{j\in\mathcal{F}} u_{c_j},\;\; \min_{j\in\mathcal{C}} u_{c_j}\,\Big], \tag{1}$$

both edges are found by bisection in $O(\log 1/\varepsilon)$ one-step evaluations, and $\mathcal{U}(x) \neq \emptyset$ exactly when $u_{\mathrm{lo}} \le u_{\mathrm{hi}}$.

Sketch. Each cap's satisfying set is a down-set and each floor's an up-set; their finite intersections are $[u_{\min}, u_{\mathrm{hi}}]$ and $[u_{\mathrm{lo}}, u_{\max}]$, whose intersection is the stated interval. (A1′) places $u_{\min}$ inside the cap family, giving the upper bisection a known-good left endpoint; $u_{\max}$ is inside the floor family if anything is, giving the lower bisection its right endpoint. Neither bisection needs the load current as a seed. □

Proposition 1 is the case $\mathcal{F} = \emptyset$. Note that (A1′) is *strictly weaker* than (A1): it asks only that zero satisfy the caps, which is the part that survives contact with a vehicle.

Remark 1 (The side of a constraint is data, not inference)*.* The plating constraint is $\varphi_{\mathrm{an}} \ge \delta$ — a lower-bound test on a signal that *decreases* in current, so it caps current from above. Sense and side coincide only when the constrained signal increases in u. Inferring side from the operator would file that cap among the floors and let the filter command straight through it: silent, and unsafe. Plants declare the side; an undeclared constraint raises rather than defaults.

Remark 2 (A soundness condition that is not obvious)*.* Delivered bus power $P = S\,V(u)\,u$ with $\mathrm{d}V/\mathrm{d}u < 0$ peaks at the maximum-power point and *falls* beyond it, so past that peak the set where the load is met stops being an up-set and the lower bisection would search a non-monotone family. The voltage cap prevents it: a pack reaches its maximum-power point near half its open-circuit voltage, far below any usable $V_{\min}$, so $u_{\mathrm{hi}} < u_{\mathrm{MPP}}$ always and the floor search is bounded by u_{hi}. This ordering is checked on every discharge platform, not assumed.

4.3 Cost

Two probes bracket the cap family, then eighteen halvings; two more bracket the floors, then eighteen more. Worst case 40 one-step model evaluations on discharge and 20 on charge, both fixed and both branch-bounded. Measured worst p99 latency across all 16 platforms is 123.9 μs (buoyancy glider), which is 1.24 % of a 10 ms ISO 26262 task slot. On the ground platforms the figure is 59.7 μs, or 0.597 %.

The property that matters is not that the number is small. It is that it does not move with the data.

4.4 The reserve

The two-sided certificate makes available a quantity the one-sided version cannot express: the width $u_{\mathrm{hi}} - u_{\mathrm{lo}}$ shrinks to zero as the envelope closes, so the filter can report *how close to unrecoverable*

Table 1: The 16 platforms. Charge-mode platforms have a zero anchor and test Proposition 1 at vehicle scale; discharge-mode platforms have a positive anchor and test Theorem 1.

Medium	**Cooling law**	**Platforms**	**Irreducible load**
Ground	Newtonian, liquid loop	robotaxi 96S45P 78 kWh; shuttle; haul truck 288S115P 601 kWh; V2G	0 (charge); 7.4–22 kW export
Air	forced air, density-derated	delivery quadrotor 6S3P; eVTOL air taxi 300S45P; turnaround	hover thrust; 400 kW; 0 (charge)
Water	sealed hull into 2 °C	survey AUV; buoyancy glider; under-ice AUV; dock recharge	25–265 W; 0.5 W; 175 W; 0 (charge)
Vacuum	Stefan–Boltzmann to 4 K (+ 6 mbar CO_2 on Mars)	LEO smallsat; GEO comsat; deep-space cruiser; Mars rover; lunar lander; 500 krad orbiter	60 W; 3 kW; 200 W; 100 W; 45 W heater; 0 (charge)

a vehicle is before it arrives. We treat this claim adversarially in §7 and scope it in §10: it carries the most operational weight and is the most likely to be overstated.

5 Method and platforms

5.1 The filter

Given a proposed input u_{req}, the filter returns $\text{clip}(u_{\text{req}}, u_{\text{lo}}, u_{\text{hi}})$, or reports the interval empty. Empty is not a violation: it is the certificate stating correctly that the irreducible load can no longer be served inside the envelope, which is the true statement and the one a mission planner needs.

Robustness enters through margins rather than estimation: a thermal throttle $\delta_T = \delta_{T0} + K(s_R - 1)$ scaled by the *assumed* resistance bound s_R, with K the smallest value certifying cells past the rated bound, and a cooling reserve f that certifies any cooling loss up to $f/(1+f)$.

5.2 One cell, sixteen vehicles

Three things vary across the platforms and nothing else does: the **medium**, through the cooling law; the **pack**, through series/parallel counts; and the **load**, through the irreducible draw. Table 1 lists them.

The cell underneath is the published reduced-order model's arithmetic in every case — same overpotential form, same plating proxy, same calibrated coefficients — with two changes the vehicles require. Current may be negative, because vehicles discharge. And the Newtonian cooling term is a callable, because a quadrotor at 400 m, an AUV at 2 °C and a spacecraft in vacuum do not share a cooling law. Under the Newtonian law with positive current the vehicle cell reproduces the published model *bit for bit*, and every result downstream inherits that.

Scope of the platform parameters. Pack geometry, hover powers, hotel loads and orbit parameters are *class-representative* — chosen to place each vehicle in the right region of its class's design space using published figures — and are not any manufacturer's design. The purpose is that "it works on autonomous vehicles" is not a testable sentence, while "it holds a 96S45P pack under a 30-minute 350 kW session at 35 °C ambient" is.

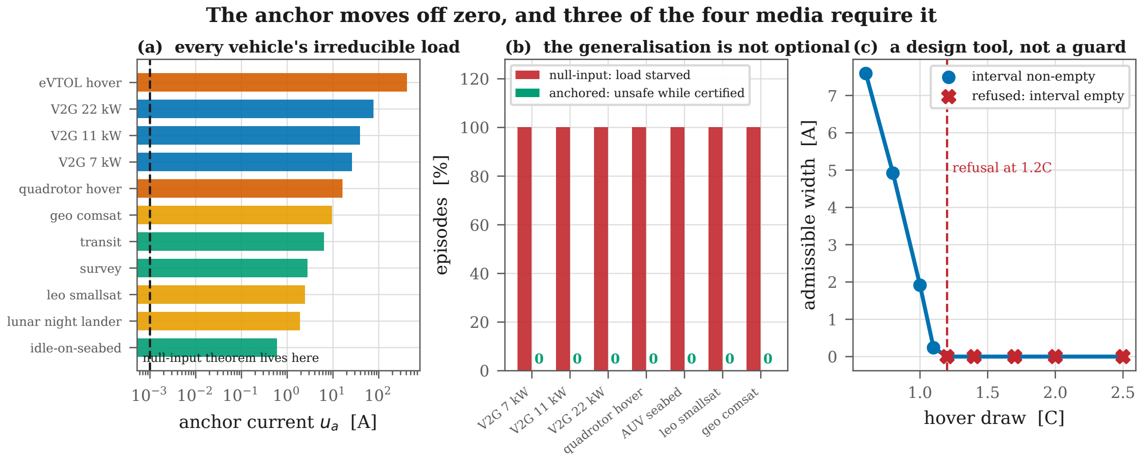

Figure 1: The anchor moves off zero, and three of the four media require it. (a) The irreducible load of every platform, spanning five orders of magnitude from a glider's 0.5 W hotel draw to an eVTOL's 400 kW hover; the null-input theorem lives at the dashed line. (b) On the same seeds, the null-input filter starves the load in essentially every episode while the anchored filter breaches nothing. (c) Applied before flight, the certificate refuses the calibrated energy cell for a rotorcraft above 1.2C of hover draw.

5.3 Experimental discipline

Every experiment runs a *pessimistic estimator against a drawn plant*: the filter is given a platform at the conservative corner of the parameter envelope, and the plant it drives is drawn from inside it. A filter tested against its own model is testing arithmetic. One property makes that corner well defined for a two-sided certificate — scaling the assumed resistance up lowers the upper edge *and* raises the lower edge, so the interval shrinks from both sides under one perturbation — and it is verified numerically, not assumed.

The experimental surface, including what would falsify each claim, was fixed in a design register before any code ran; divergences between the register and what happened are recorded rather than edited into agreement. Zero-failure claims are reported with Clopper–Pearson 95 % one-sided upper bounds [21]; continuous outcomes with bootstrap confidence intervals; paired comparisons with Wilcoxon signed-rank and effect sizes; monotonicity claims with permutation Spearman.

6 Ground: robotaxis, shuttles, autonomous trucks

The ground domain is where the original result should hold most directly — the anchor is zero, the medium is a liquid loop, the physics is what the model was calibrated on. What is new is scale and duty.

6.1 The filter is unchanged, at pack scale

On one cell the vehicle filter is bit-exact with the published projection: 0 mismatches in 100,000 states, across both discretisations, fresh and aged. A P-parallel pack's admissible current is exactly P times one cell's, to a relative deviation of 6.5×10^{-16}. And on a 33,120-cell haul-truck pack the admissible current equals the minimum over its heterogeneous cells to $1.9e - 13$ A — below the $4.3e - 13$ A resolution of the search itself. A pack-averaged thermal limit on that many cells is a statement about a cell that does not exist; the weakest-cell result is what makes a per-cell certificate tractable.

6.2 Megawatt charging and the weakest cell

The haul truck is the scale case, and it is where the per-cell architecture stops being a design preference. A 288S115P pack is 33,120 cells with individually different resistances, capacities and plating margins. A pack-averaged thermal limit on that many cells is a statement about a cell that does not exist — the average cell is not the one that fails.

The per-cell alternative is only tractable because of a structural result: the pack's admissible current is exactly the minimum over its cells'. We verify this from $N = 1$ to $N = 33,120$ with heterogeneous ageing, and the agreement is $1.9e-13$ A — below the $4.3e-13$ A resolution of the search itself, which is to say exact. The consequence is that a fleet operator does not need 33,120 filters; one filter evaluated at the weakest cell's parameters is provably the pack's answer.

Megawatt charging makes this urgent rather than academic. The binding constraint at that power is not the connector or the cable; it is the coldest, most-aged cell in a 600 kWh pack, and identifying which cell that is by measurement is not something a truck stop can do.

6.3 A 350 kW session

Under the full parameter envelope, 4,000 sessions on a 78 kWh robotaxi pack at 35 °C produced 0 violations (Clopper–Pearson 95 % upper bound 0.075 %), with a worst plant temperature of 39.9 °C against a 45 °C limit, and the pessimistic corner dominating every draw. Note which constraint binds: 350 kW into a 96S pack is about 985 A, above the pack's own 3C ceiling of 540 A. **The charger is not the limiting element; the certificate is.**

Over five years of robotaxi duty — 21,915 fast-charge sessions with the filter given a fixed datasheet bound on day one and told nothing afterwards — there were 0 violations (CP-95 0.0137 %). At twelve DC fast charges a day against a private car's one a fortnight, a factor of 171, a robotaxi accumulates a private car's *decade* of fast-charge stress in 3.0 weeks.

6.4 The result this domain was not looking for

Coupled straight to ambient air, the throttled margin puts the admissible ceiling at $T_{\max} - \delta_T = 32.5$ř$C$. Above roughly that ambient there is *no* admissible fast charge at all, and the certificate delivers nothing — correctly, because at the datasheet end-of-life resistance bound there is none to deliver. The measured dead zone begins at 35 °C, the first grid node past the prediction.

Running the identical sweep with a loop holding coolant at 25 °C recovers the whole range and buys **+8.7 points of delivered charge** on average, with 0 violations in all 8,000 sessions across both configurations. That difference is exactly what a thermal management system purchases, quantified, and it is why production packs carry chillers rather than radiators.

A smaller finding follows. Delivered charge is not monotone in ambient and was never going to be: it peaks at 25 °C, rising along the cold limb ($\rho = +0.884$) where series resistance limits and falling along the hot limb ($\rho = -0.661$) where the thermal margin does. A *binary* service metric — "will it make 60 % in thirty minutes" — jumps the full range between adjacent grid cells, because thresholding a smooth quantity produces a threshold. Dispatch should plan on the curve the filter already computes, not on the binary derived from it.

6.5 What the certificate is worth to a fleet

These quantities translate into fleet operations directly, and the mechanism is the clearest case in this paper of a safety result having an economic consequence.

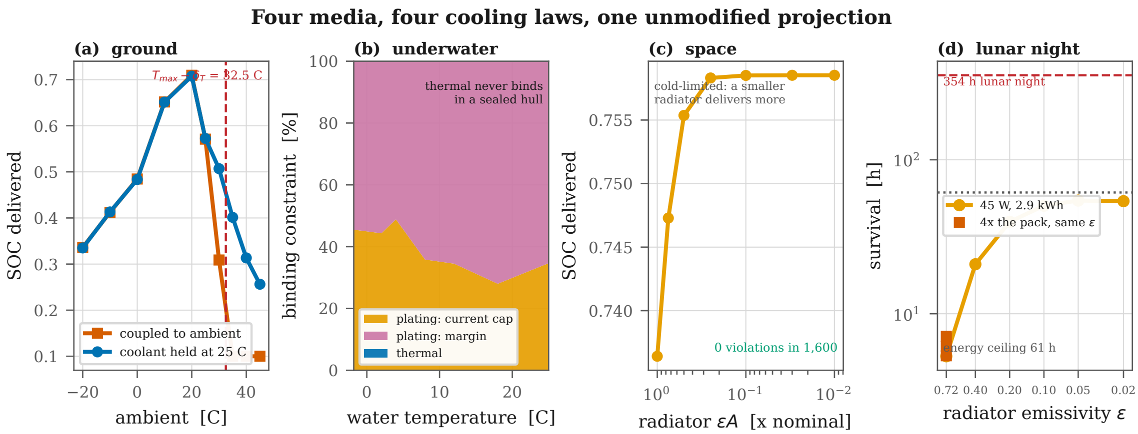

Figure 2: Four media, four cooling laws, one unmodified projection. (a) Ground: the passive ceiling at $T_{\max} - \delta_T$ and what a coolant loop recovers. (b) Underwater: plating binds at every water temperature and thermal never does; what shifts with cold is the share taken by the temperature-dependent current cap. (c) Space: delivered charge *rises* as the radiator shrinks, because the pack is cold-limited. (d) Lunar night: capacity barely moves survival, insulation moves it by an order of magnitude, until the energy ceiling takes over.

A production pack ships with a de-rate chosen for a worst case: end of life, temperature extreme, weakest cell. That worst case is almost never the present case. A vehicle that is warm, mid-life and mid-charge is being held to a limit designed for a vehicle that is none of those things. The certificate replaces the fixed de-rate with an evaluation at the *current* state under a bound that is still conservative — nothing is given up in safety — and the difference is recovered continuously.

The 8.7-point gain from active cooling above is one instance made measurable. So is the ambient curve: a dispatcher who knows delivered charge peaks at 25 °C and falls along each limb for a different physical reason can schedule charging around weather rather than against it. And the stress-equivalence result sharpens the whole question — a robotaxi is not a car that drives more, it is a vehicle whose battery ages at roughly 171 times the rate its de-rate was designed around. A guarantee holding for 21,915 sessions with no recalibration is what makes that duty cycle defensible rather than merely profitable.

6.6 Sensing, transients, and vehicle-to-grid

A thermistor reading low is survivable up to a breakpoint with a closed form, $b^* = \delta_T/(1 - \Delta t\, hA/C_{\mathrm{th}}) = 12.802\,\check{r}C$, using the throttled margin actually in force. The measured breakpoint is 12.9 °C on a 0.1 °C grid — an error of **0.098 °C**, from published constants with nothing fitted. Regenerative-braking pulses at 10 Hz over 1,500 episodes produced zero certified violations, and a coolant-degradation sweep over 8,000 trials violated nothing anywhere inside the region the cooling reserve predicts.

Finally, vehicle-to-grid gives the ground domain its own instance of the generalisation, which matters because it removes the objection that anchored collapse is a story about aircraft. At 7.4, 11 and 22 kW of contracted export — anchors up to 76.4 A — the anchored filter recorded zero unsafe episodes in 600 sessions each, while the null-input filter dropped the contracted load in **600 of 600** at every power.

7 Air: delivery rotorcraft and eVTOL

This is the domain the generalisation exists for, and the one where the null-input filter does not merely under-perform but loses the aircraft. On identical seeds, 600 flights each: the anchored filter recorded 0 unsafe-while-certified episodes; the null-input filter starved the rotors in **600 of 600**, serving a mean of 0.0 steps before brownout. It does not fly the aircraft for a while and then fail. It never flies it at all.

The two-sided admissible set is genuinely an interval: 20,000 flight states scanned densely returned 0 disconnected sets, with every bisected edge within one grid step of the dense scan. Across the whole flight envelope — payload × altitude, gusts to $\sigma = 0.5$ (3,000 flights), cold soak at 2 000 m, motor-out to +100 % power (3,000 flights) — nothing was breached while the certificate reported feasible.

7.1 The certificate refuses to fly the calibrated cell

The most interesting aerial result is a refusal. With the model's own calibrated coefficients — an energy cell — the anchored filter reports the interval **empty before takeoff** at and above 1.2C of hover draw. The admissible width collapses smoothly toward it: 5.06 A at 0.6C, 1.28 A at 1.0C, 0.16 A at 1.1C, empty at 1.2C (Fig. 1c).

This is not a filter failure. It is the filter working as a *design tool.* The correct engineering answer to "may I fly a rotorcraft on an energy cell" is no, and here that answer arrives before the aircraft leaves the ground rather than as a thermal event in flight. A runtime guard is not normally able to produce it, because a runtime guard is only ever asked about the vehicle it is already installed in.

The remaining aerial experiments therefore use a power cell — 0.40× resistance, 0.60× capacity — which is the largest single modelling liberty in this work and is flagged wherever it applies.

7.2 The reserve warns, and the takeoff transient sizes the pack

Across 1,200 flights, *every* closure preceded its breach: 0 breaches with no prior warning, median lead 24 s, 5th percentile 16 s, 95th percentile 62 s. Under gusts the result holds at every variance up to $\sigma = 0.5$ with 0 unsafe episodes in 3,000 flights — once the *live* closure is distinguished from a debounced one. Requiring three consecutive empty steps before declaring closure costs up to 96 s at the highest gust variance, and measuring lead from the debounced signal made an earlier version of this experiment report *negative* lead times for a filter that had already closed correctly.

A full delivery sortie — takeoff, climb, cruise, hover over the drop, return, land — shows the reserve tracking the mission without the filter being told a mission plan exists: 3.07 A in takeoff, 16.61 A in climb, 31.22 A in cruise, 26.23 A over the drop (Fig. 3).

Takeoff is the binding phase, and it is what sizes the pack. On 6S2P the certificate refuses a 1.75× takeoff transient outright; on 6S3P it allows it and 34.2 % of sorties complete; on 6S4P, **78.5 %**, with 0 unsafe episodes throughout. The frontier is actionable, which is what separates a design tool from a complaint.

Payload and density altitude close the envelope through the same physical term — momentum theory places both inside $T^{3/2}/\sqrt{\rho A}$ — and the reserve falls with each: $\rho = -0.981$ against payload and $\rho = -0.998$ against hover power, the shared term, and negative against altitude in *every* payload stratum. Pooled across the grid the altitude correlation is a non-significant -0.192 ($p = 0.359$), because payload spans a far larger range of hover power and swamps it; the within-stratum test is the one that answers the question.

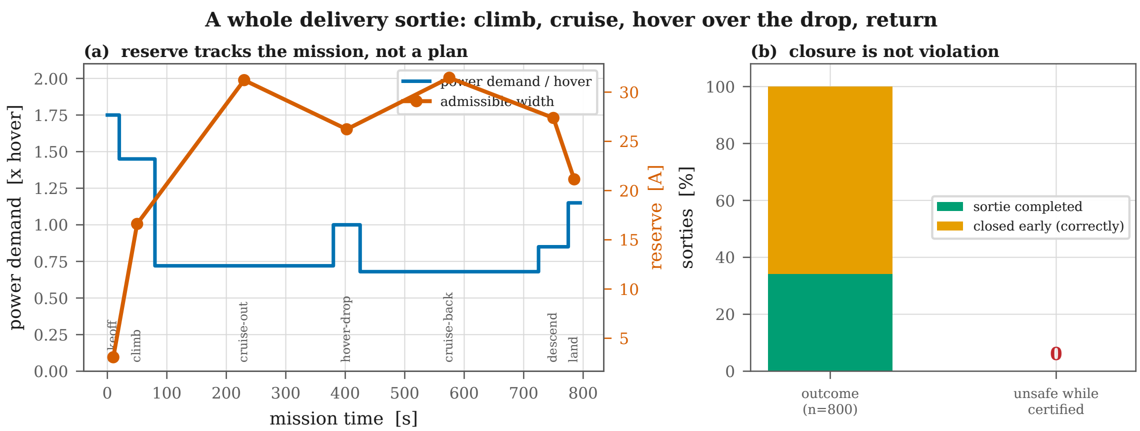

Figure 3: A whole delivery sortie. The reserve opens in cruise, where power demand is low and forced-air cooling is high, and closes over the drop. The filter is never told a mission plan exists; it sees a load that changes and re-derives both edges every step. Closure is not violation: the amber fraction is the certificate correctly refusing to continue.

Table 2: Binding edge for a delivery quadrotor at 2 000 m density altitude, 400 flights per node. Warm, the actuator ceiling limits and the aircraft is thrust-limited. Cold, series resistance climbs and *voltage sag* limits — and the interval closes entirely below -20 řC. Nothing was breached while certified at any node.

Air temp.	Mean reserve	Voltage	Thermal	Actuator	Closed
-30 řC	2.09 A	228	0	0	172
-20 řC	7.11 A	298	0	0	102
-10 řC	13.64 A	354	0	0	46
0 řC	20.20 A	279	0	120	1
$+10$ řC	25.30 A	108	0	292	0
$+20$ řC	27.35 A	45	0	355	0
$+30$ řC	28.01 A	9	0	391	0
$+40$ řC	17.73 A	0	3	251	146

An eVTOL air taxi hovering at 3.09C certifies with zero unsafe episodes, a median hover endurance of 3.0 minutes, and interval width predicting that endurance at $\rho = +0.694$.

7.3 Which edge binds, and why cold is the aerial problem

The two-sided certificate makes it possible to ask which constraint is actually limiting, and the answer moves with temperature in a way that is not intuitive. Table 2 reports the binding edge for a quadrotor at 2 000 m across an 70 K air-temperature range.

Two things follow. First, the thermal constraint — the one an aviation engineer would expect to dominate a high-power pack — essentially never binds; *voltage sag* does, everywhere below freezing. Second, the envelope closes outright in 43 % of states at -30 řC: a cold-soaked delivery drone at altitude is not marginally degraded, it is refused. Only at $+40$ řC does thermal appear at all, and even there it accounts for 3 of 400 states.

An operator planning winter service on the basis of thermal derating would be planning against the wrong constraint. The certificate identifies the right one without being told which to look for, because it evaluates all of them.

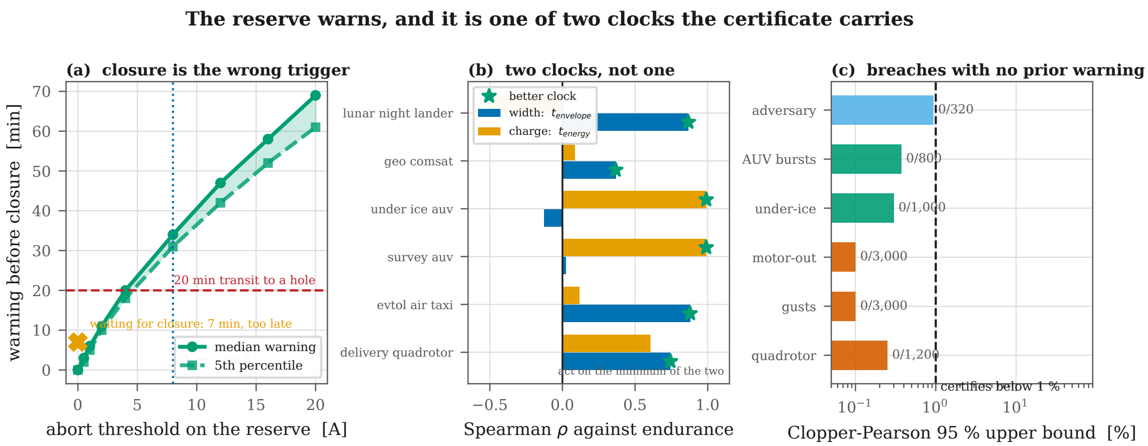

Figure 4: The reserve warns, and it is one of two clocks. (a) Under ice, closure itself leaves only 7 minutes before a breach against a 20-minute transit; aborting on a reserve threshold gives warning ahead of closure, and 8 A covers the transit. (b) Interval width predicts endurance where the envelope is what closes and does not where the pack simply empties; the certificate carries both clocks. (c) Breaches with no prior warning, across every stress: all zero.

7.4 What forty sorties a day actually costs

A delivery pack lands at up to 44 °C, and the throttled margin puts the admissible ceiling at 32.5 °C — so the certificate refuses to charge it. Measuring how little charge a refused session delivers measures the refusal. The operational question is how long the pack must sit, and the filter answers it directly, because *the wait is the time until the interval opens*.

Of 4,000 turnarounds, 18 % could start at once; the rest waited a median of **60.8 minutes** (95th percentile 100.9), and none was still refused after two hours, with zero violations throughout. On a single pack that is 14.5 sorties/day. Swapping packs so the cool-downs overlap gives 37.2 sorties/day, and holding a 40/day target requires **2.8 packs in rotation**.

This is why delivery operators swap packs rather than wait. The contribution is that the certificate produces the number instead of the folklore.

7.5 An adversary

A cross-entropy adversary was given 320 independently drawn packs and 30,720 flight profiles, free to command anywhere inside the certified interval and optimising the demand and airspeed schedule to maximise peak temperature. It reached 79.2 % of the limit while the certificate was open and broke nothing: 0 violations, CP-95 upper bound 0.932 %, past the 299 cells needed to certify below 1 %. The statistical unit is the *cell*, not the sequence, because an earlier adversarial claim in this line of work rested on 40 cells and bounded nothing below 7.2 %.

8 Water: survey AUVs, gliders, under-ice vehicles

The hardest domain, for reasons that have nothing to do with the theorem. A pack inside a pressure hull is thermally isolated from an excellent heat sink — 0.0069 W/K per cell against a car's 0.0799, about one 12th — and the water is 2 °C, so both intuitions are wrong at once: it heats slowly *and* it is very cold.

It is also the cleanest separation of the anchored generalisation from anything about flight. A

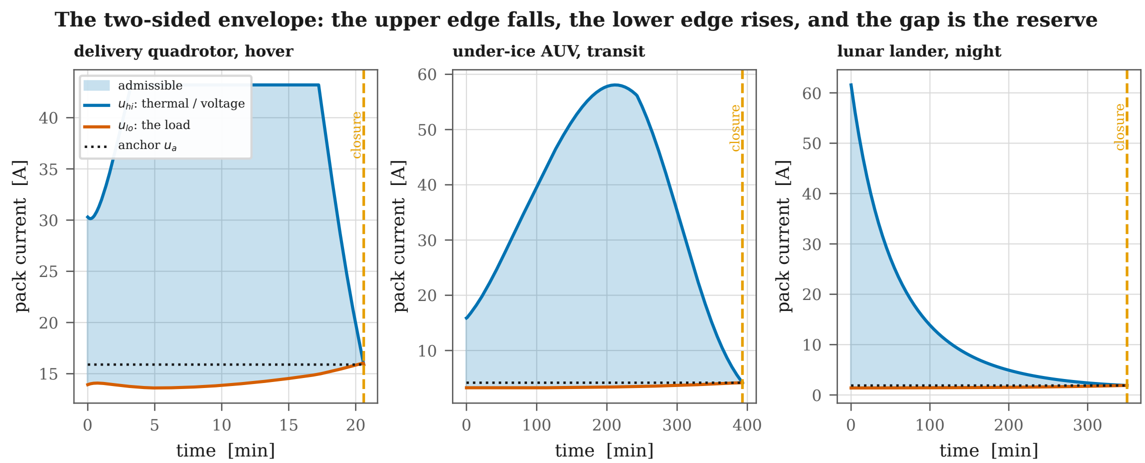

Figure 5: The two-sided envelope in three vehicles. The upper edge falls as the pack heats and sags; the lower edge rises as the load costs more current at lower terminal voltage; the gap between them is the reserve, and closure is where they meet. The dotted line is the anchor — the current the irreducible load demands, which is where the lower edge lands. On a charging cell that line sits at zero and the whole lower half of this picture is absent.

survey AUV sitting perfectly still on the seabed still cannot command zero: its computer, inertial navigation and acoustic modem draw 25 W regardless, an anchor of 0.60 A. Across idle, survey and transit the anchored filter recorded zero unsafe-while-certified episodes, and the null-input filter browned out the vehicle in **800 of 800 — including the case where nothing is moving.**

A sealed hull certifies: 3,000 dock recharges in 2 °C water, 0 violations. A six-month deployment with no recalibration: 547 recharges, zero violations. And 180 days with no ground truth: a fixed bound and a daily oracle record identical safety *and* identical delivered charge.

8.1 The constraint the theorem does not certify is the one that binds

This experiment was written to show the binding constraint *switching* from thermal to plating as the water cools. It never switches, because it was never thermal. **Plating binds in 100 % of states at every water temperature from** -1.8 **to 25 °C, and temperature binds in none** (Fig. 2b), across 4,200 trials. A sealed hull charges so slowly that the pack never approaches its thermal limit. What changes with temperature is which plating mechanism binds: the temperature-dependent current cap takes 35 % of cases at 25 °C and 44 % at 2 °C.

That has a consequence worth stating plainly rather than dressing up. The certified set is $\{T, V\}$; plating is enforced and monitored but never certified, because $u = 0$ does not clear the plating margin at high state of charge and so no one-step condition can make it invariant. **On this vehicle, in this medium, the binding constraint is the one the theorem does not certify.**

Two downstream sweeps inherit it and both come back flat. Delivered charge does not move across ten years of dormancy (spread $2.0e-08$ SOC) or 6 000 m of depth ($7.6e-08$), because neither channel touches the plating cap. Those spreads are numerical noise and no rank correlation is reported over them — an earlier version did, and reported $\rho = +1.000$, $p = 0.0002$ for a flat line. *The certificate is not tolerant of dormancy and depth so much as blind to them.*

Table 3: Under-ice abort rule. Two different quantities are involved and conflating them would overstate the result. Waiting for closure gives *no* warning before closure by definition — it *is* closure — and leaves only 7 minutes before an actual constraint breach. Aborting on a threshold in the reserve gives warning *ahead* of closure, which is the quantity a transit needs. Only the two-sided certificate has a reserve to threshold.

Abort trigger	Median warning before closure	5th percentile	Covers 20-min transit
wait for closure	0 min (by definition)	0 min	0 %
2 A	11 min	10 min	0 %
4 A	20 min	18 min	54 %
8 A	**34 min**	**31 min**	**100 %**
16 A	58 min	52 min	100 %

8.2 A negative result, and the design rule it forces

Under an ice shelf there is no abort-to-surface: a closed envelope is a lost vehicle. The question is therefore not whether the certificate is correct but whether it is *early*.

Waiting for the interval to close is too late. Closure is the moment the envelope is gone, so it offers no warning ahead of itself; what it does offer is the interval before the vehicle actually breaks something, and across 1,000 missions that is a median of 7 minutes. Against a 20-minute transit to a known hole in the ice, it covers the return in 0 % of episodes. 0 breaches occurred without a warning — the certificate is correct — and correct 7 minutes before the damage, when the vehicle needs 20, loses the vehicle anyway.

The fix is not a better certificate. It is to stop using closure as the trigger. The reserve is available at every step and falls smoothly toward zero, so the abort can fire on a *threshold* rather than on exhaustion (Table 3). Aborting at 8 A of remaining reserve gives 34 minutes of warning *before* closure at the median and 31 at the 5th percentile — covering the transit in every episode, with the 7-minute post-closure margin still in hand on top of it.

It is worth stating what this rule replaces. Current practice for autonomous underwater endurance is a fixed reserve fraction — turn back at 30 % state of charge, say — chosen conservatively because the quantity that actually matters, how much admissible current remains, is not observable. A fixed SOC reserve is simultaneously too conservative in warm water with a fresh pack and not conservative enough in cold water with an aged one, and nothing tells the vehicle which situation it is in.

The reserve threshold is directly observable, updates every control step at 123.9 μs, and carries the temperature and ageing state implicitly because it is computed from them. That is the difference between a mission rule that is defensible and one that is merely traditional.

8.3 Four decades of load, one certificate

A buoyancy glider drawing 0.5 W and an under-ice AUV drawing 175 W differ by a factor of 350 in load. Both produced single-interval admissible sets in every state scanned and zero unsafe episodes. What differs is the reserve *ratio* — 1575 for the glider against 7.7 for the AUV — which is the dimensionless statement of how much room each has relative to what it must draw.

9 Vacuum: LEO, GEO, deep space, Mars, lunar night

Space is where the certificate's least glamorous property becomes the decisive one. The filter requires no identification of the plant: no recursive least squares, no observer to converge, no ground

Table 4: Five mission classes and what each one makes hard. The certificate is unmodified across all of them; only the cooling law, the pack and the irreducible load change.

Mission class	Defining stress	What it makes hard	Measured result
LEO smallsat	5,662 cycles/year, no technician	count, not severity	28,307 charges, 0 violations
GEO comsat	72-min eclipse, payload cannot go quiet	depth of discharge at end of life	every duration completes, 0.0 % DoD
Deep-space cruise	4–250 K sink, hours of light time	recalibration is impossible	oracle buys 0.00 SOC points
Mars surface	80 K diurnal swing, 6 mbar CO_2	two cooling laws at once	2,400 trials, 0 violations
Lunar night	354 h at 100 K, heater is the load	both edges push on temperature	survival frontier measured, not passed

station in the loop. On a car that is a convenience. On a spacecraft that has not been touched since integration, whose ground link is hours of light time away, and whose battery has taken 500 krad of ionising dose in the interim, **it is the only reason a fixed pre-launch parameter set is admissible at all.**

Space is also where the *form* of the guarantee matters as much as its content. Flight software is qualified by argument about worst-case behaviour, and an iterative solver is difficult to argue about: its run time depends on data that has not happened yet, its convergence depends on conditioning nobody can bound in advance, and its failure mode is to return late rather than to return wrong. A projection that always executes the same bounded sequence of arithmetic operations, allocates nothing, and branches a fixed number of times is a different kind of object to qualify. It is the difference between arguing that a component usually finishes and proving that it always does.

9.1 Four environments

Low Earth orbit is defined by its count rather than its severity: at a 92.9-minute period, 5,662 charge cycles a year. Five years — 28,307 charges, never recalibrated — produced 0 violations (CP-95 0.0106 %), with delivered charge on the final quarter matching the first to three decimal places.

Geostationary orbit is the opposite duty cycle: about ninety eclipses a year clustered into two seasons, the longest 72 minutes, with a payload that is not allowed to go quiet. Every duration out to the seasonal maximum completed, at a median depth of discharge of 0.0 %, with zero unsafe episodes.

Deep-space cruise radiates to a 4–250 K sink and nothing else: 3,000 draws, 0 violations, with the Stefan–Boltzmann cooling channel verified monotone in temperature numerically rather than assumed. **The Martian surface** adds 6 mbar of CO_2 to radiation across an 80 K diurnal swing: 2,400 trials at hourly resolution, 0 violations, with the sum of the two cooling terms confirmed monotone.

Ionising dose enters as one more monotone channel: zero violations to 500 krad, with delivered charge falling gently and monotonically in dose.

9.2 Why the count is the hard part in low Earth orbit

The LEO number is easy to misread as routine. Fifteen and a half charge cycles a day for five years is 28,307 cycles executed by a filter given its parameters before launch and sent nothing since. Over that span the pack's series resistance grows, its capacity fades and its plating margin tightens — and the certificate does not know any of it. It does not need to, because a monotone *bound* suffices where a point estimate would be required, and the bound was known at integration.

The observable consequence is that delivered charge on the last quarter of the mission matches the first to three decimal places. An operator's usual expectation is that battery performance is a slowly degrading resource requiring periodic re-planning. Here it is flat — not because the pack has not aged, but because the certificate was never relying on the pack being new.

9.3 Two inversions of terrestrial intuition

Losing the radiator helps. Sweeping εA down by two orders of magnitude produced 0 violations at every point — and delivered charge *rose*, from 0.736 to 0.759 (Fig. 2c). On a car, losing cooling costs charge. Radiating to a 4–250 K sink, a healthy radiator makes the pack *cold*, series resistance climbs, and the voltage limit arrives sooner; shrinking it lets the pack keep its own dissipation. In this domain the pack is cold-limited, not heat-limited, and a thermal-control failure of this kind is not an emergency.

Lunar night is a cold problem, so capacity is the wrong lever. A lander on 2.9 kWh survives about 5.3 hours of a 354-hour night, and quadrupling the pack buys only a couple more — because the envelope closes on cold-induced voltage sag, not on stored energy. Insulation is the lever: dropping emissivity from 0.72 to 0.05 takes survival from 5.3 to **54.3 hours**, at which point the pack stops being cold-limited and meets its energy ceiling of 61.3 hours. No configuration survives the full night, which is the honest answer and the reason real landers that survive lunar night carry radioisotope heaters rather than bigger batteries.

These two results share a structure worth naming. Terrestrial battery engineering is organised around removing heat, because on Earth the ambient sits close to the cell's operating temperature and the failure mode is thermal. In vacuum the ambient is 4 K and the failure mode inverts: the pack's problem is *retaining* heat, series resistance is the mechanism, and voltage sag is how it presents. A method tuned to terrestrial intuition — one that treats cooling as unambiguously good — would prescribe exactly the wrong design for both the deep-space probe and the lunar lander. The certificate makes no such assumption. It evaluates the constraints it is given under the cooling law it is given, and the inversion falls out.

9.4 No ground in the loop

Over 1,825 charges — five years — a fixed datasheet bound and an oracle recalibrated every single day record identical safety *and* identical delivered charge: the gap is 0.00 SOC points. In deep space, recalibration buys not just no safety but no performance either, because the binding constraint is the plating corner rather than resistance.

Solar-array degradation, swept to 60 years — three times any real mission, down to 29.8 % of beginning-of-life output — never becomes the binding constraint at all, and delivered charge varies by 0.0009 SOC across the whole sweep. The separation between what the vehicle can *request* and what the filter can *certify* is not marginal here.

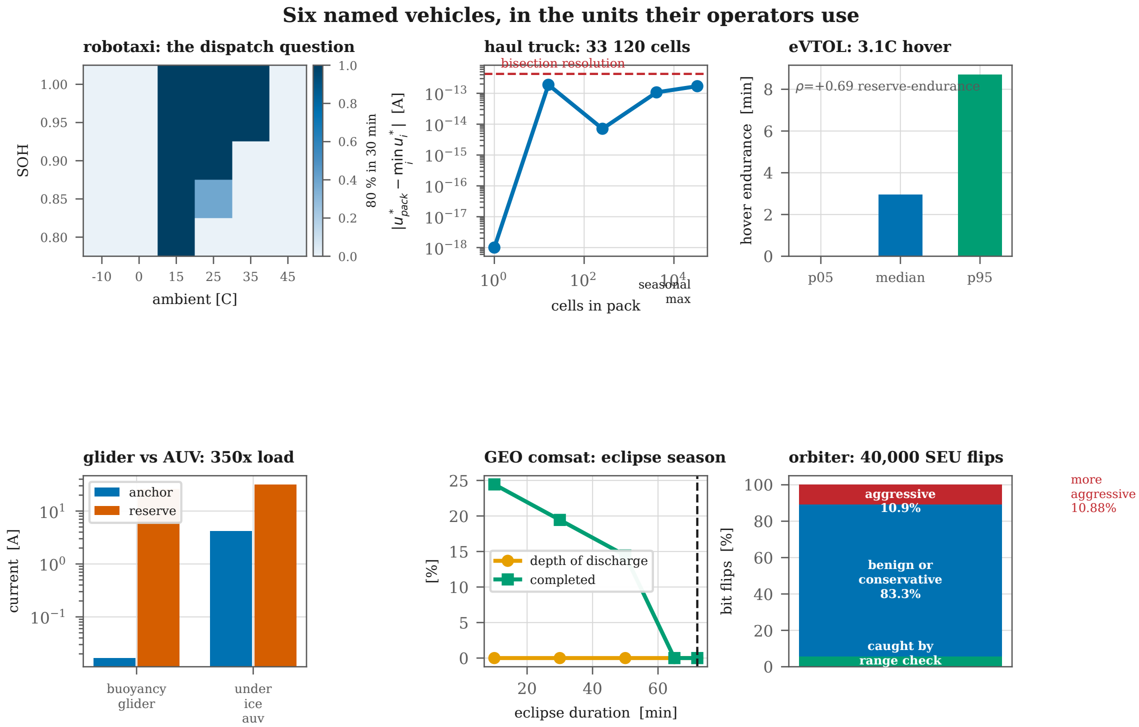

Figure 6: Six named vehicles, in the units their operators use: robotaxi dispatch feasibility over state of health and ambient; the weakest-cell result at 33,120 cells; eVTOL hover endurance; a glider and an AUV four decades of load apart; a GEO eclipse season; and single-event-upset containment on a high-radiation orbiter.

9.5 The one fault the theorem cannot survive, measured anyway

Every other fault model in this work corrupts the plant or the sensor. A single-event upset corrupts the *certificate*: a heavy ion flips a bit in the state word the projection is about to read, so the filter computes an admissible interval for a spacecraft that does not exist and commands it to the one that does.

No one-step theorem survives corruption of its own input, and claiming otherwise would be dishonest. What can be measured is the shape of the exposure. Across 40,000 bit flips into state of charge, temperature and the RC overpotential: 5.80 % are caught by a three-comparison range check on the state word; 83.31 % are benign or leave the command *more* conservative; and 10.88 % produce a more aggressive command than the clean one. Of those, the worst overshoot actually realised in the plant was **0.000 °C** — none breached anything in one step. The deliverable is the range check and the number beside it, not a theorem.

10 What makes this one method

Forty-four experiments establish that the certificate holds in each medium. That is not the same as establishing it is the *same* certificate — four domains that each worked for their own reasons would be four papers. Five cross-cutting checks, each of which would fail if the domains were only superficially related.

The admissible set is a single interval everywhere. 64,000 dense scans across all 16 platforms and both modes: 45,570 single intervals, 18,430 empty, **0 disconnected**. The two-bisection

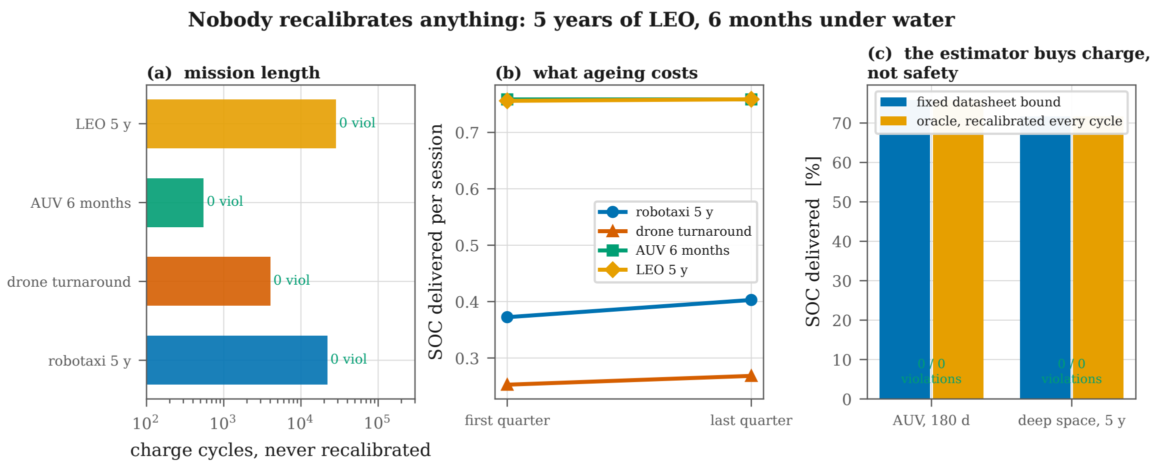

Figure 7: Nobody recalibrates anything. (a) Mission length by domain, on a log scale, with violations annotated. (b) What ageing costs across each mission: delivered charge per session, first quarter against last. (c) A fixed datasheet bound against an oracle told the plant's true resistance before every cycle — identical safety, and on these two missions identical delivered charge.

Table 5: Spearman ρ against measured endurance for the two quantities the certificate carries. Every platform is well predicted by one of them; none by both.

Platform	**Width clock** $u_{\text{hi}} - u_{\text{lo}}$	**Charge clock** $(\text{soc} - \text{soc}_{\text{min}})Q/u_{\text{lo}}$
eVTOL air taxi	+**0.879**	+0.115
lunar-night lander	+**0.867**	-0.503
delivery quadrotor	+**0.744**	+0.606
GEO comsat	+**0.367**	+0.084
survey AUV	+0.023	+**0.990**
under-ice AUV	-0.124	+**0.990**

construction rests entirely on this, and it is measured rather than argued from monotonicity.

The generalisation reduces to the original theorem exactly. Bit-exact on 30,000 states with 0 mismatches, and on every charge-mode platform the lower edge is exactly zero. Proposition 1 is recovered, not approximated.

The cost is fixed, and it is the same fixed cost everywhere. 20 evaluations on every charge platform and 40 on every discharge platform, against theoretical bounds of exactly 20 and 40. Worst p99 latency across all sixteen is 123.9 μs, or 1.24 % of a 10 ms task slot.

Every zero-failure claim is backed by enough samples. Twelve headline claims, 69,389 pooled trials, 0 violations; all exceed the 299 trials needed to certify below 1 % at 95 % confidence and five exceed the 2,995 for 0.1 %.

10.1 The reserve is one of two clocks

The claim that interval width is a countdown survives in the air and on the Moon and does *not* survive underwater, and pooling would have hidden it (Table 5).

An interval can close for two different reasons. *Envelope-driven* closure is the upper edge falling onto the lower one as the pack heats and sags — there the width *is* the distance still to travel. *Energy-driven* closure is the state of charge reaching its floor with the interval still wide open — there the clock is set by how much charge was in the pack, and the width has nothing to say about

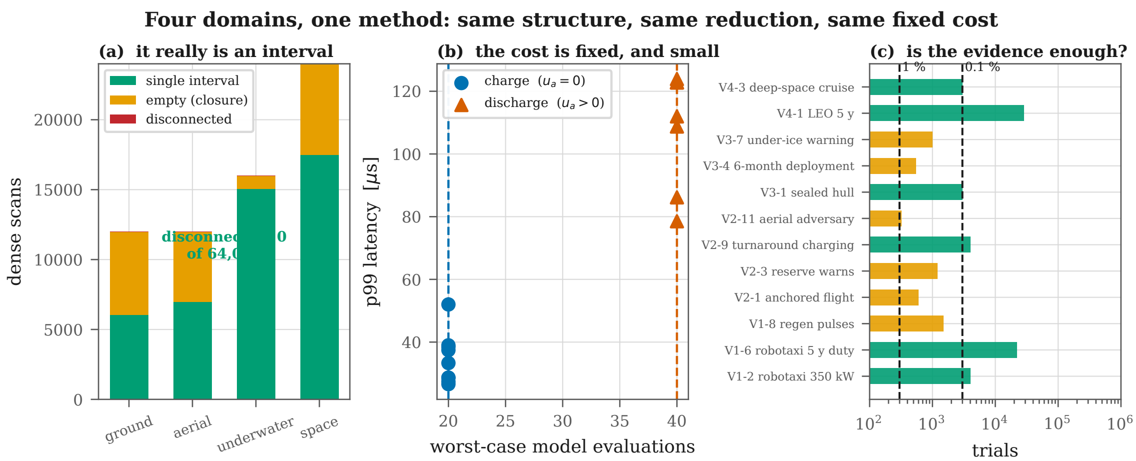

Figure 8: Four domains, one method. (a) Every dense scan returns a single interval or an empty one; none is disconnected. (b) The evaluation count is bounded at 20 on charge and 40 on discharge, and latency follows it. (c) Every headline claim clears the sample size needed to certify below 1 %.

it. Both are correctly reported as closures; only the first is a countdown.

The certificate already carries both quantities, and every platform has one of them significantly positive ($\rho > 0.3$, $p < 0.001$), the geo comsat most weakly at $+0.367$. **The operational rule is to act on the minimum of the two**, which is conservative whether or not either is individually tight. The pooled, anchor-normalised correlation is $+0.916$ over $2,400$ episodes; reporting that alone would have been the most flattering presentation available and the least honest.

11 What this enables

The results above are safety results; their significance is what they permit. What follows separates what has been demonstrated from what follows from it.

Fleets can charge at the physical limit rather than a fixed de-rate. The de-rate a production pack ships with is chosen for a worst case that is almost never the present case; evaluating at the *current* state under a still-conservative bound recovers the difference continuously. §6 quantifies what a thermal-management system buys at 8.7 points of delivered charge — otherwise a design decision made on intuition.

Aircraft can be sized by a certificate rather than by flight testing. The takeoff frontier of §7 is a design curve: which pack configurations can certify the transient the aircraft must actually fly, before an airframe is built. The refusal of §7.1 rejects a cell chemistry for a mission class in one evaluation.

Vehicles that cannot be rescued can be given an abort rule with a number in it. §8.2 converts “turn back when you are low” into “abort at 8 A of remaining reserve, which gives 31 minutes of warning at the 5th percentile”. That is a rule a mission planner can defend, and it exists only because the two-sided certificate produces a reserve.

Spacecraft can fly a fixed parameter set for a decade and know what it costs. §9 measures that cost directly against a daily oracle and finds it to be 0.00 SOC points — that is, nothing. For a mission that cannot recalibrate, this is the difference between a guarantee and a hope.

And the guarantee is cheap enough to be everywhere. 123.9 μs of deterministic, branch-bounded computation on a scalar model fits inside any microcontroller a battery management system already contains. There is no solver to certify, no library to qualify, and no failure mode in which the computation does not finish.

12 Limitations

Stated in full, because several of them are load-bearing.

No high-fidelity electrochemistry in this study. Every number here is reduced-order-model level or model-versus-model; the Doyle–Fuller–Newman evidence validating the underlying filter is prior work, cited rather than re-derived.

The certified set is $\{T, V\}$; plating is enforced, never certified. §8 shows this is not a technicality: in a sealed hull at any water temperature, the binding constraint is the one the theorem does not certify. Extending the certificate to plating would require a control-invariance argument that the $u = 0$ backup cannot supply, and we do not have one.

One cell, sixteen vehicles. Every platform uses the same calibrated coefficients, because that is the cell the model was fitted to. Where a vehicle class would really use a different cell — the rotorcraft — the substitution is explicit, is 0.40× resistance and 0.60× capacity, and is an extrapolation beyond the calibration. Every aerial number carries it.

Class-representative platforms. Pack geometry, hover powers, hotel loads and orbit parameters place each vehicle in the right region of its class's design space; they are not any manufacturer's design.

No theorem survives corruption of its own input. §9 measures the single-event-upset exposure and offers a range check; it does not claim robustness to a flipped bit, because there is none to claim.

Simulation of simulation. The plants are models, and a filter tested against models chosen to satisfy its hypotheses is weaker than one tested against hardware. What partially answers this is that three of the four media *break* the original hypothesis, one platform is refused outright, and four experiments contradicted the hypothesis they were written to test. The register was not constructed so that everything passes. Hardware validation is the natural next step and is not claimed here.

13 Conclusion

The safety guarantee an autonomous vehicle's battery needs turns out to require neither an optimiser nor an identified model, only a geometric fact: when constraints are monotone in a scalar input and one input is known safe, the admissible set is an interval, and an interval can be found by bisection at fixed cost.

Getting that from a cell to a vehicle required noticing that the theorem's apparent failure in flight was a misreading. Zero current is still safe for every temperature and voltage constraint on a discharging pack; what zero cannot do is serve the load the vehicle exists to serve, and that is a floor rather than a cap. Anchored Collapse adds the floor family and a second bisection, and nothing else — 40 fixed evaluations against the original 20.

Across 16 platforms in four media and 49 experiments totalling 470,923 episodes, nothing was breached while the certificate reported feasible, the admissible set was a single interval in all 64,000 dense scans, and the worst latency anywhere was 123.9 μs.

The results we would most want a reader to take are the ones that did not come out as intended. A certificate that refuses a cell chemistry for a mission class before takeoff is doing something a runtime guard is not normally asked to do. A domain in which the binding constraint is the one the theorem explicitly does not certify is a limitation a less careful register would not have surfaced. And a warning that leaves 7 minutes when the vehicle needs 20 is a negative result whose forced abort rule — act on a threshold in the reserve, not on its exhaustion — is more useful than the positive result would have been.

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